Jay Chao

OPT/H-1B | willing to relocate | 669-600-1942 | yjaw.job@gmail.com | linkedin.com/in/yjaw | github.com/yjaw

Education

University of Wisconsin-Madison Dec 2025
Master of Science (M.S.) in Computer Science *WI, USA*

National Tsing Hua University Jan 2022
Post-Baccalaureate in Computer Science *Hsinchu, Taiwan*

Tunghai University Jul 2018
Bachelor of Business Administration (B.B.A.) *Taichung, Taiwan*

Skills

Programming Languages: Python, C++, C#, Golang, Java, JavaScript, HTML/CSS, SQL
Frameworks: FastAPI, .NET, Spring, Node.js, React, Ray, OpenAI API
Tools: Kubernetes, Docker, KubeRay, Argo CD, Git, GitHub Actions (CI/CD), Linux, RESTful APIs, gRPC, k6, LLM
Databases & Cloud: MySQL, PostgreSQL, SQL Server, Azure, AWS

Experience

NeuraSignal – Software Engineer (Contract) Oct 2026 – Present
Ultrasound Simulation Platform – *Python, C++, NumPy*

- Developing collision and stochastic Doppler simulators in Python and C++ for a research-grade medical ultrasound simulation platform on a 4-engineer team, spanning DSP, numerical validation, and automated testing

Open Source – Contributor Jan 2026 – Present
KubeRay & Ray – *Go, Kubernetes, GitHub Actions*

- Traced up to **55s** of extra startup time per new Ray worker on Kubernetes to a fixed init-container limit; opened a fix with unit tests that sizes it from the Ray container
- Fixed History Server requests failing for suspended RayClusters by keying off the controller's suspended condition, not the spec flag, to avoid racing the log collector's final flush
- Reduced manual CI work: merged a flaky e2e-test fix and a stale issue/PR bot, and opened a `/retest` workflow that reruns failed GitHub Actions and Buildkite jobs from a PR comment without ever running PR code

Apache Airflow – *Python, TypeScript, React*

- Merged fixes across the React web UI and provider packages, including a rendering bug fix and a declarative YAML migration of connection metadata
- Wrote a locale skill for Airflow's AI translation agent, defining terminology and style rules for LLM output

IBM – Software Engineer Intern Jul 2025 – Aug 2025
Cloud-Native Modernization – *C++, gRPC, Docker, Kubernetes, Argo CD, Linux, k6*

- Built a bridge that exposes a closed-source legacy CORBA service over gRPC so Kubernetes can health-check it, automating a previously manual validation workflow (**2x faster**)
- Cut deployment time from about **30 to 5** minutes by replacing manual deployments with GitOps releases on Kubernetes (Argo CD), supporting Blue-Green deployments
- Built a k6 gRPC load-testing suite simulating **1,000** concurrent users to find where latency spiked; presented the prototype and performance findings in a live demo

Projects

Mini ERP: Purchasing & Inventory System – *C#, ASP.NET Core, SQL Server, Azure, OpenAI API* 2026

- Built a procurement system covering request, tiered approval, purchase order, partial receiving, and inventory, with controls that block self-approval and conflicting concurrent edits
- Made the supplier on-time-delivery report **5x faster** (992 ms → 197 ms on 100K requisitions) by tracing the slowdown to a per-row time-zone conversion in a SQL view
- Added natural-language queries with the OpenAI API, read-only by design: generated SQL runs as a database user that can only read approved views